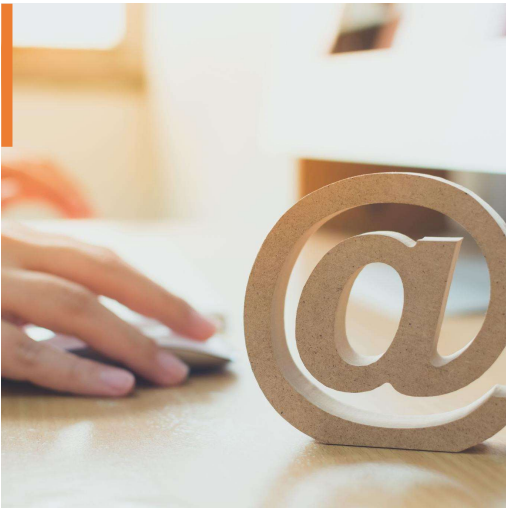




MGT 3213 - Management Information Systems

The Database Approach to Data Management

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Data Vs. Information

Example:

Warner	10021
Miller	10022
Jones	10023
Ward	10024

Example:

<u>Employee_name</u>	<u>Employee_ID</u>
Warner	10021
Miller	10022
Jones	10023
Ward	10024

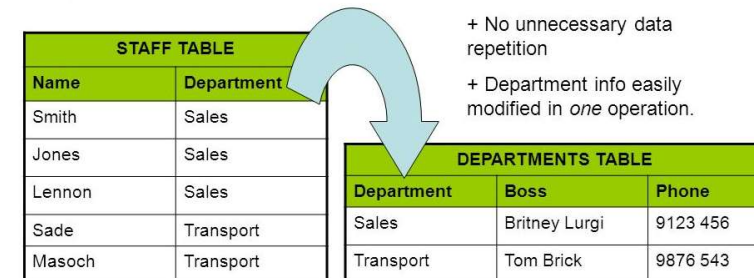
Flat File

Name	Department	Boss	Phone
Smith	Sales	Britney Lurgi	9123 456
Jones	Sales	Britney Lurgi	9123 456
Lennon	Sales	Britney Lurgi	9123 456
Sade	Transport	Tom Brick	9876 543
Masoch	Transport	Tom Brick	9876 543

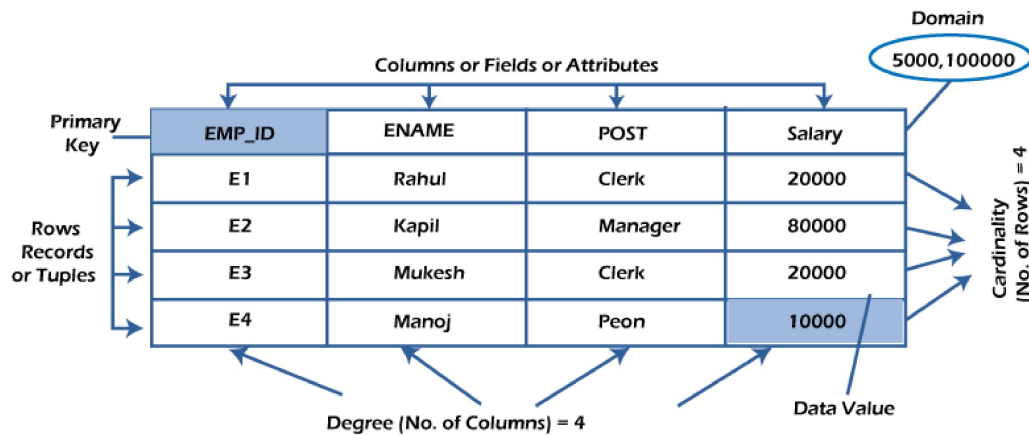
Flat File vs. Relational Database

- A flat file is usually a table with rows of information, whereas a relational database contains many tables with rows and columns that recognize relationships between different pieces of information

Name	Department	Boss	Phone
Smith	Sales	Britney Lurgi	9123 456
Jones	Sales	Britney Lurgi	9123 456
Lennon	Sales	Britney Lurgi	9123 456
Sade	Transport	Tom Brick	9876 543
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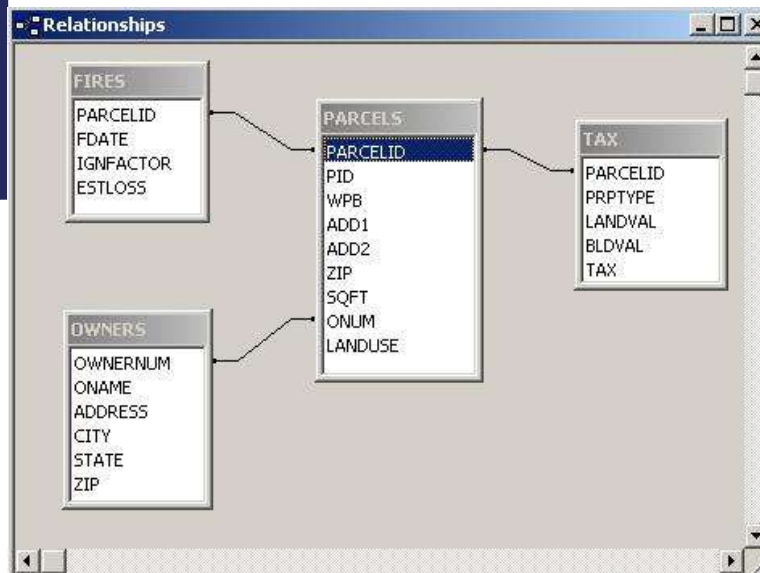
What is RDBMS



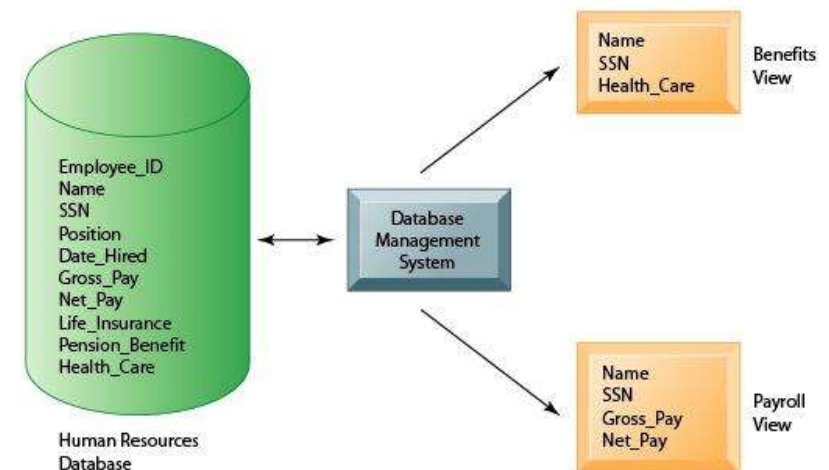
Database Technology

- Database technology eliminates many of the problems of traditional file organization by organizing data:
 - centralizing data
 - controlling redundant data
 - serve many applications and different groups at the same time.

Data Modelling - Relationships



Human resources database with multiple views



Database management system (DBMS)

- A database management system (DBMS) is software that:
 - Acts as an interface between application programs and the data files
 - Separates the logical view of the database (how the data is perceived by end users) and the physical view (how the data is actually organized on storage media).
 - It also allows different logical views for different users.
 - Helps to reduce data redundancy and eliminate data inconsistency by allowing a central, shared data source



Managing data

Data modelling

- Data are being designed through the different models, the relationship between the data and other details are portrayed through this concept.

Data mining

- It is used for transforming raw data into information.
- It has wide use in Industries.

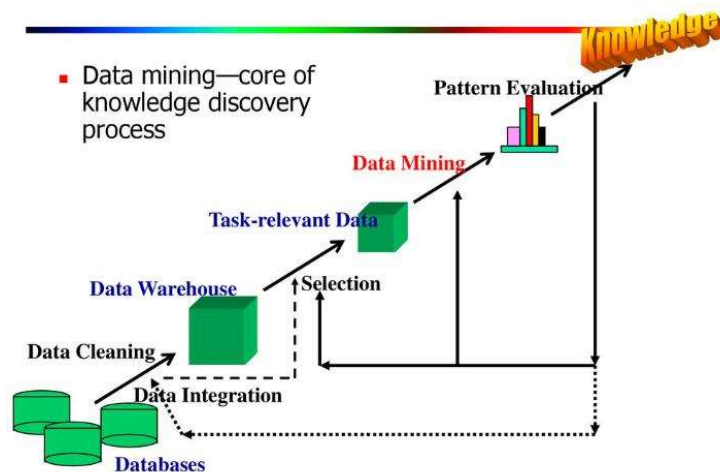
Data integration

- It combines different data from different sources.
- It also analyzes those data for the processing of information.

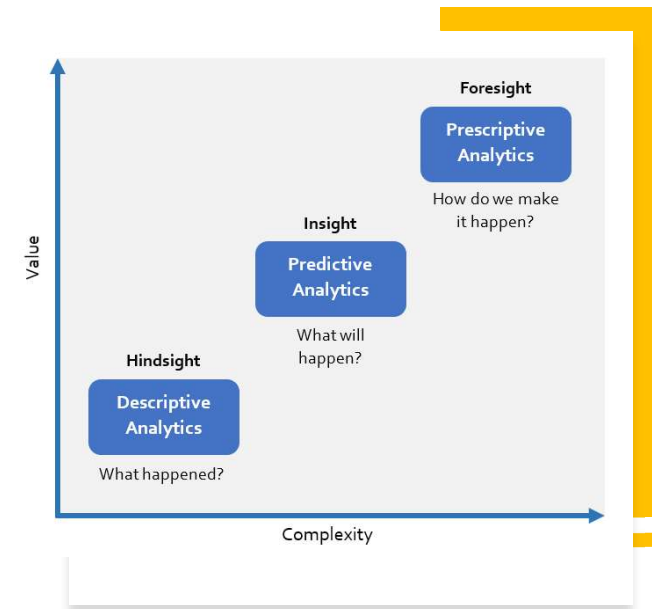
Data governance

- Data handling policies are made under this concept.
- It also confirms data fetching consistency and other related issues.

Knowledge Discovery Process



Descriptive, Predictive, and Prescriptive Analytics



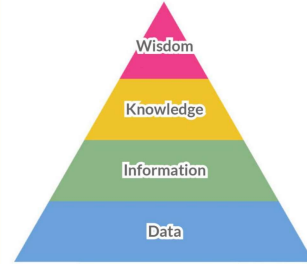
WHAT IS DATA MINING?

A set of techniques to extract valuable information from large data volume

Apply statistics or artificial Intelligence to extract valuable information from large database

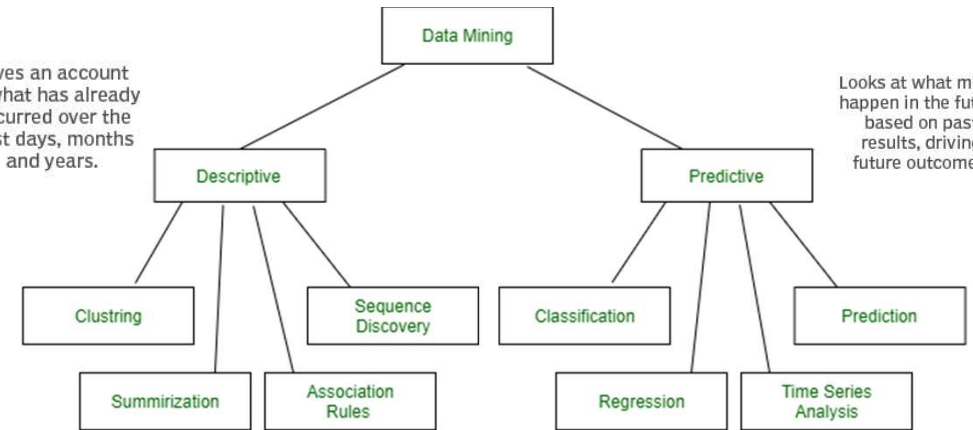


The 4 categories information gathered from Data mining



Data Mining Models

Gives an account of what has already occurred over the past days, months and years.



Looks at what might happen in the future based on past results, driving future outcomes.

References

- [https://paginas.fe.up.pt/~als/mis10e/ch6/chpt6-2bullettext.htm#:~:text=A%20database%20management%20system%20\(DBMS,actually%20organized%20on%20storage%20media\).](https://paginas.fe.up.pt/~als/mis10e/ch6/chpt6-2bullettext.htm#:~:text=A%20database%20management%20system%20(DBMS,actually%20organized%20on%20storage%20media).)
- <https://www.geeksforgeeks.org/data-mining-process/>

The End